Roedean Tour de Maths 14 May 2015 Leg 4 Total: 160 Marks

- Answer all Questions on your answer sheet (Just the answers, not your working).
- Rough paper is given for working.
- Calculators may be used (CASIO only☺).
- May the force be with you.

CASIO®

ROEDEAN SCHOOL (SA)
INSPIRING A LIFE OF SIGNIFICANCE

5 MARK QUESTIONS

1. Determine the next number in the sequence:

   $0; 1; 9; 36; 100;$ ___ (5)

2. When the members of a marching band are arranged in rows of 2, 3 or 4 there is always one person left. However, when they are arranged in rows of 5, there is no one left over.
   What is the minimum number of people in the band? (5)

3. What number should replace the question mark below?

6	**3**	**4**	**6**
5	**5**	**7**	**4**
8	**3**	**4**	**8**
3	**9**	**7**	**6**
7	**7**	**8**	**?**

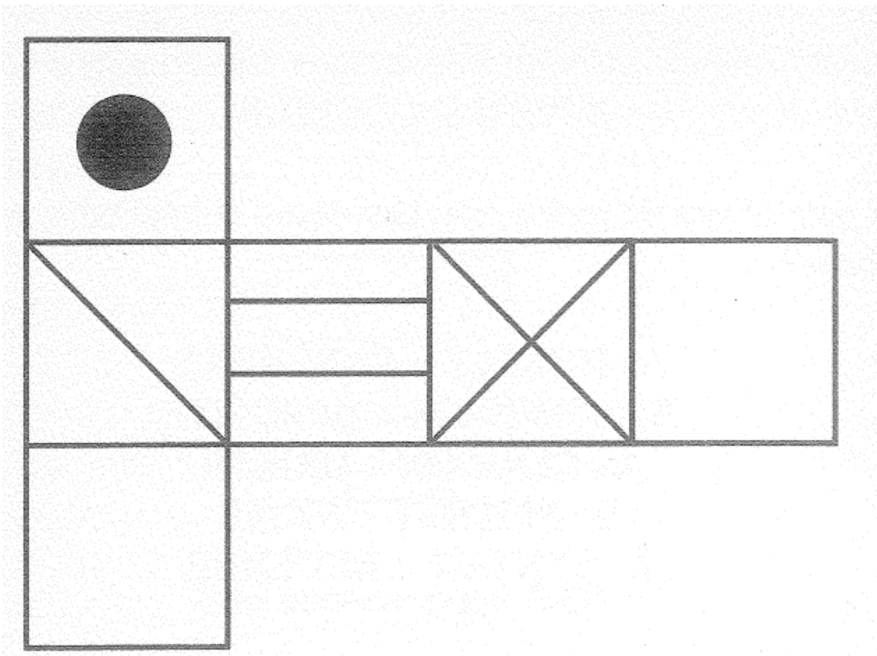
(5)

4. What number should replace the question mark below?

5 4	6 7	8 4	9 3	8 1
27	**40**	**71**	**34**	**?**
7 6	9 7	5 9	5 4	7 7

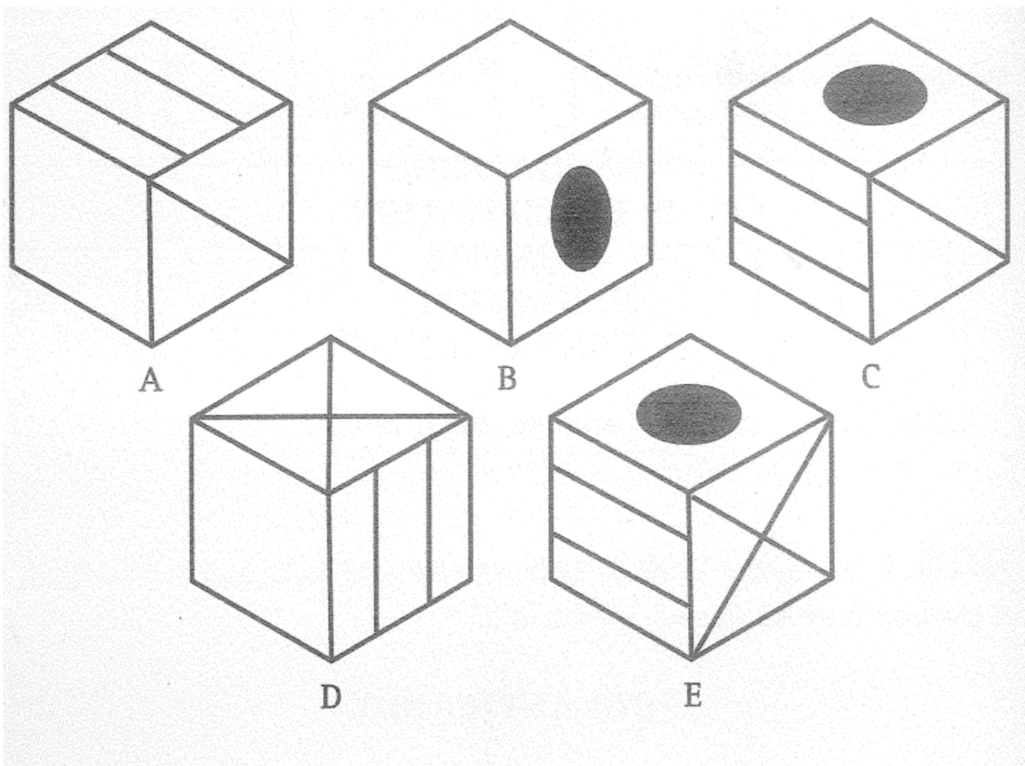
(5)

5.



When the above is folded to form a cube, just one of the following can be produced.

Which one is it?



6. A farmer with $240m$ of fencing wishes to enclose a rectangular area of the greatest possible size.
What will be the greatest area surrounded?
Give your answer in $metres\ squared$. (5)

7. Determine the value of $3^{4000} + 3^{4000} + 3^{4000}$.

Write your answer in exponential form. (5)

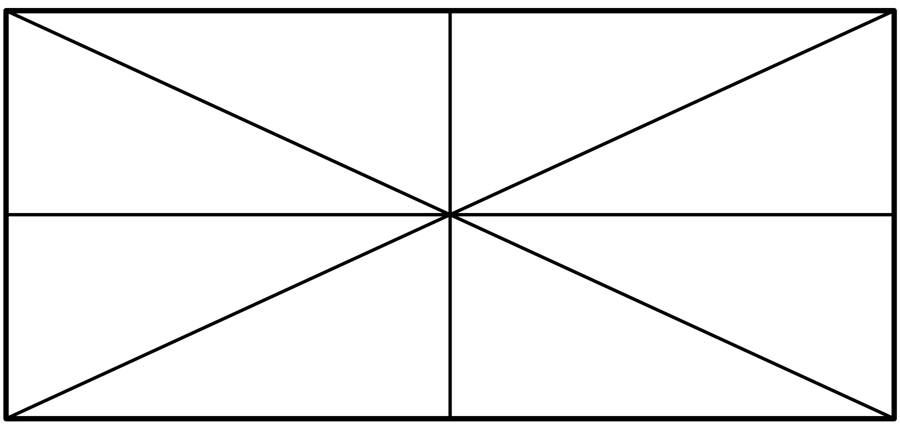
8. If we count backwards in sevens from 77003 which of number(s) in the fifties will be counted. (5)

9. A survey of 50 boys' participation in summer sport revealed that:

- 28 boys participated in swimming.
- 19 boys participated in cricket.
- 10 boys participated in neither.

How many boys participated in cricket but didn't participate in swimming? (5)

10. How many triangles are there in the figure below?



(5)

10 MARK QUESTIONS

11. Rose and David throw one die each.

What is the probability (as a fraction in simplest form) that Rose throws a higher number than David? (10)

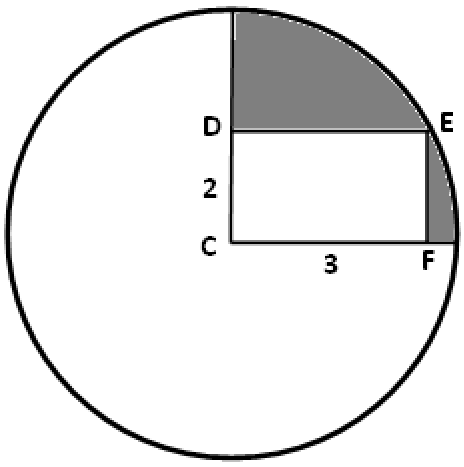
12. A train, $250m$ long, travelling at a speed of $40km/h$, enters a tunnel that is $2{,}25km$ long.
How long does it take for all of the train to pass through the tunnel from the moment the front enters it, to the moment the rear emerges?
Give you answer in minutes and seconds. (10)

13. A company employs 64 females and 36 males.

The average absenteeism is $6\frac{1}{4}\%$ for females and $8\frac{1}{3}\%$ for males.

What is the average absenteeism overall? (10)

14.

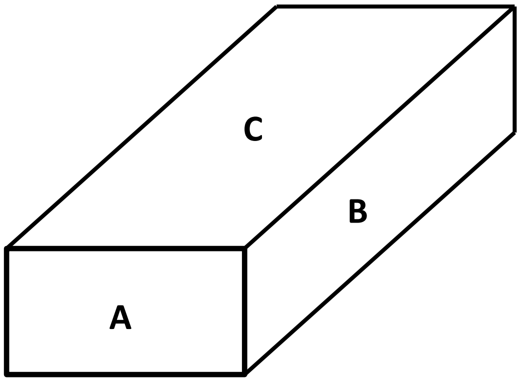


In the diagram above:

- C is the centre of the circle.
- E is a point on the circle such that CDEF is a $2cm \times 3cm$ rectangle.

Determine the area of the shaded region in cm^2 to 2 decimal digits. (10)

15.



The following information is given about the rectangular prism above:

- It has integer valued dimensions.
- The area of face A is $24cm^2$.
- The area of face B is $40cm^2$.
- The volume of the prism is $240cm^3$.

Determine the area of face C in cm^2. (10)

20 MARK QUESTIONS

16. The following happened at a party:

- By 8:00pm, all the guests had arrived.
- By 8:30pm, one-third of the guests had left.
- By 9:30pm, one third of the guests who were at the party at 8:30pm had left.
- By 11:00pm, one third of the guests who were at the party at 9:30pm had left.
- By 11:00pm there were 16 guests left at the party.

How many guests were at the party at 8:00pm? (20)

17. Out of 100 people surveyed:

- 86 had an egg for breakfast.
- 75 had bacon for breakfast.
- 62 had toast for breakfast.
- 82 had coffee for breakfast.

How many people, *at least*, **must** have had all four items for breakfast? (20)

18. Find the value of:

$$\frac{1}{1\times 2}-\frac{1}{2\times 3}-\frac{1}{3\times 4}-\frac{1}{4\times 5}-\ \ldots\ldots\ldots\ldots.-\frac{1}{98\times 99}-\frac{1}{99\times 100}$$

(20)

THE END